field is the region around the magnet where its magnetic influence can be felt. It is denoted by B and its unit is Tesla.

The direction of the magnetic field around a magnet can be found by placing a small compass in the magnetic field (Fig 5.1).



Figure 5.1 Compass showing direction of magnetic field

Magnetic field can penetrate through all kinds of materials, not just air. The Earth produces its own magnetic field, which shields the earth's ozone layer from the solar wind and it is important for navigation also.

5.2 Magnetic Field Lines

A magnetic field line is defined as a curve drawn in the magnetic field in such a way that the tangent to the curve at any point gives the direction of the magnetic field. They start at north pole and ends at south pole. In Figure 5.2, the arrow mark indicates the direction of magnetic field at points A, B and C. Note carefully that the magnetic field at a point is tangential to the magnetic field lines.

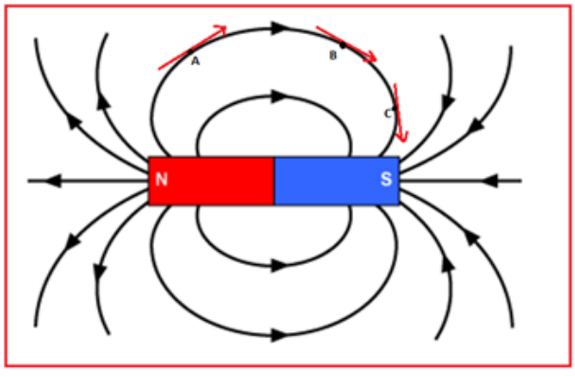


Figure 5.2 Magnetic field lines

5.2.1 Magnetic flux

Magnetic flux is the number of magnetic field lines passing through a given area (Fig. 5.3). It is denoted by ϕ and its unit is weber (Wb).

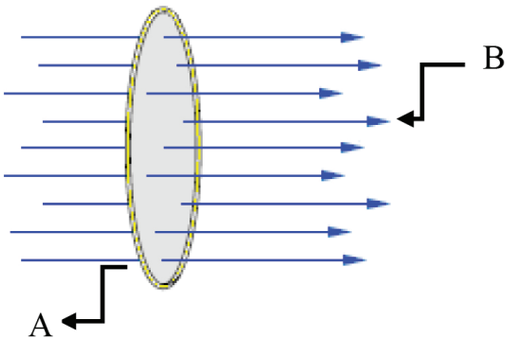


Figure 5.3 Magnetic flux

The number of magnetic field lines crossing unit area kept normal to the direction of field lines is called magnetic flux density. It is shown in Figure 5.4. Its unit is Wb/m^2

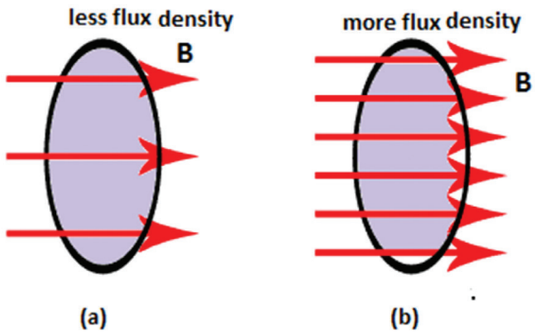


Figure 5.4 Magnetic flux density

DO YOU KNOW?

Some sea turtles (loggerhead sea turtle) return to their birth beach many decades after they were born, to nest and lay eggs. In a research, it is suggested that the turtles can perceive variations in magnetic parameters of Earth such as magnetic field intensity and remember them. This memory is what helps them in returning to their homeland.



IX_SCI_EM_Unit-05_PHY.indd 53 30/11/2022 19:22:56